Aula 2

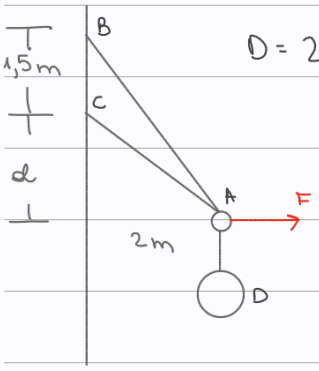
05 / 03 / 26

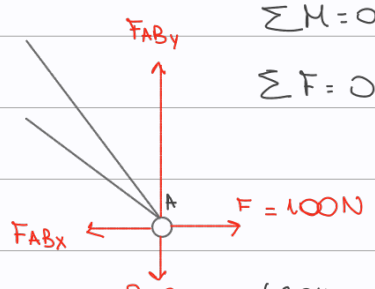
- momento é uma força vezes uma alavanca.
- Momento torção e flestor ; tangencial e ortogonal

Aula 3. Condição de equilíbrio 10/03/2026

$F = 100N$

$D = 20kg$





$P = 200N$ $(20kg \cdot 10)$ → gravidade

$\Sigma M = 0$

$\Sigma F = 0 \rightarrow \Sigma F_x = 0 \quad \Sigma F_y = 0$

$-F_{ABx} + F = 0$ $\quad F_{ABy} - 200 = 0$ $\quad F_{AB} = \sqrt{F_{ABy}^2 + F_{ABx}^2}$

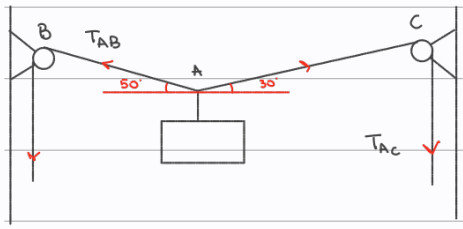
$F_{ABx} = 100$ $\quad F_{ABy} = 200$ $\quad F_{AB} = 223,6$

$$\begin{cases} F_{AB} \operatorname{sen}\theta = 200 \\ F_{AB} \cos\theta = 100 \end{cases}$$

$F_{AB} = \frac{100}{\cos\theta} \rightarrow 100 \cdot \frac{1}{\cos\theta} \rightarrow 100 \cdot \frac{1}{\cos\theta} \cdot \operatorname{sen}\theta = 200$

$\frac{\operatorname{sen}\theta}{\cos\theta} = \frac{\cancel{200}}{\cancel{100}} \rightarrow tg\theta = 2$

Questão 3. "Dois operários descarregam um caixote de madeira de massa igual a 75 Kg através de um sistema de polias. Determine as forças que os dois operários exercem nos respectivos cabos.



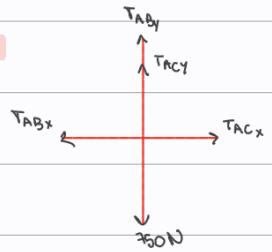
$\Sigma F_x = 0$

$-T_{ABx} + T_{ACx} = 0 \rightarrow T_{ABx} = T_{ACx}$

$\Sigma F_y = 0$

$T_{ABy} + T_{ACy} - 750 = 0$

$T_{ABy} + T_{ACy} = 750$



$T_{ACx} = T_{AC} \cos 30°$

$T_{ACy} = T_{AC} \operatorname{sen} 30°$

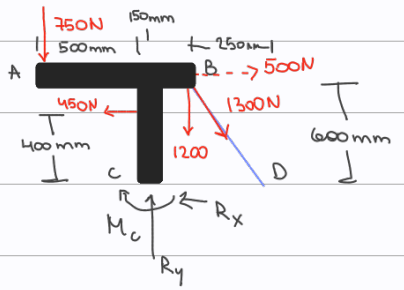
$T_{ABx} = T_{AB} \cdot \cos 50°$

$T_{ABy} = T_{AB} \cdot \operatorname{sen} 50°$

$T_{ABx} = T_{ACx} \rightarrow T_{AB} \cdot \cos 50 = T_{AC} \cdot \cos 30 \rightarrow T_{AB} = T_{AC} \dfrac{\cos 30}{\cos 50}$

$T_{ABy} + T_{ACy} = 750 \rightarrow T_{AB} \operatorname{sen} 50 + T_{AC} \operatorname{sen} 30 = 750 \rightarrow T_{AC} =$

Questão 4. "Determine as reações em C sabendo que a força no cabo BD é igual a 1300 N".



1300 · cos

$\Sigma F_x = 0$

$-450 + 500 - R_x = 0$

$R_x = +50$

$\Sigma F_y = 0$

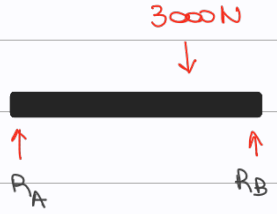
$-750 - 1200 + R_y = 0 \rightarrow R_y = 1950\,N$

$\Sigma M_c = 0$ (foi escolhido o Ponto C!)

$(+750 \cdot 500mm) + (450 \cdot 400) + M_c - (1200 \cdot 150) - (500 \cdot 600) = 0$

$M_c = +75\,Nm$ (só mudar o sentido na ilustração).

Questão 5. "Determine as reações de apoio em A e B na viga bi-apoiada ao lado".



$\Sigma F_y = 0$

EQ1

$R_A + R_B - 3000 = 0$

$R_A + 2000 = 3000 \rightarrow R_A = 1000\,N$

$\Sigma M_A = 0$

EQ2

$0 - (3000 \cdot 2,4) + (R_B \cdot 3,6) = 0$

$R_B = 2000\,N$

Momento no Ponto A.

Momento no Ponto B

$\Sigma M_B = 0$

EQ2

$-R_A \cdot 3,6 + (3000 \cdot 1,2) = 0$

$R_A = 1000\,N$

$\Sigma M_C = 0$

EQ2 EQ3

$-R_A \cdot 2,4 + R_B \cdot 1,2 = 0$

Momento no Ponto C

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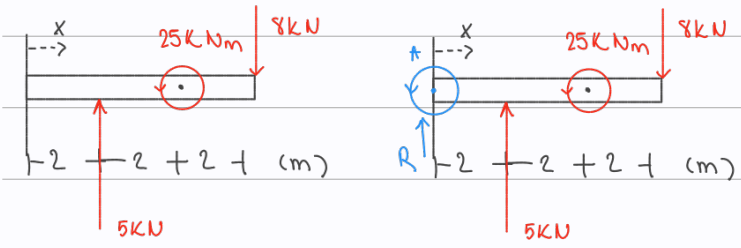
Questão 6. "Determine as reações de apoio na viga engastada ao lado".



$\Sigma F_y = 0$

$R_x - 8K + 5K = 0 \rightarrow R_x = 3KN$

$\Sigma M = 0$

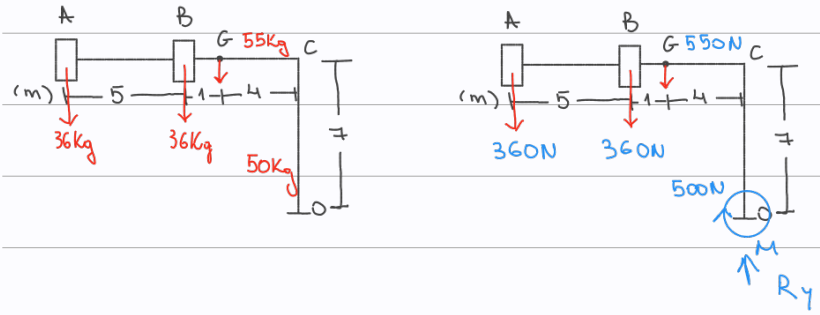
$M_A + (5K \cdot 2) + (25K) - (8K \cdot 6) = 0$

$M_A + 35KNm - 48KNm = 0$

$M_A = 13 KNm$

Provável prova:

Questão 7 "Calcule as forças e o momento na base O do poste mostrado ao lado sabendo que as massas dos elementos AC e OC são 55Kg e 50Kg respectivamente e cada semáforo tem massa igual a 36Kg".



$\Sigma F_y = 0$

$-360 - 360 - 550 - 500 + R_y = 0$

$R_y = 1770 N$

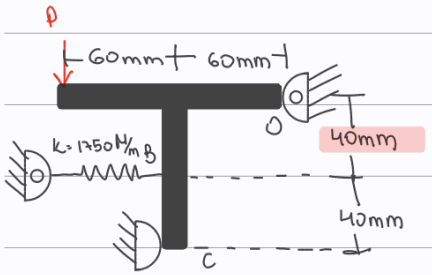
$\Sigma M_o = 0$

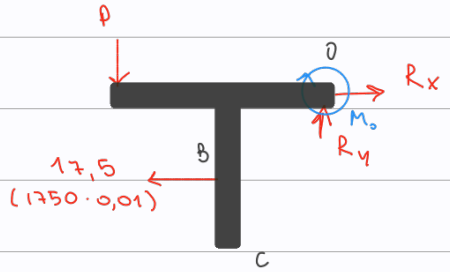
$+ (360 \cdot 10) + (360 \cdot 5) + (550 \cdot 4) + \underbrace{(500 \cdot 0)}_{+} - M_o = 0$

$M_o =$

Provável prova!

Questão 8. "Calcule a força P necessária para interromper o contato em C sabendo que a mola está esticada em 10 m".





$\Sigma F_x = 0$

$R_x - 17,5 = 0 \rightarrow R_x = 17,5\ N$

$\Sigma F_y = 0$

$-P + R_y = 0 \rightarrow R_y = P = 5,83\ N$

$\Sigma M_o = 0$

$+(P \cdot 120mm) - (17,5 \cdot 40\ mm) - 0 = 0$

$P = \dfrac{17,5\ N \cdot 4\ \cancel{mm}}{12\ \cancel{mm}} \rightarrow 5,83\ N$ //